

MS4215: Lab 4

Inference: Wages and Work Experience

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Introduction

The dataset `CPS1985` comes from the 1985 Current Population Survey: a large-scale survey of the US workforce conducted by the Bureau of Labor Statistics. It contains data on 534 workers across a range of occupations and industries. The variables used in this lab are:

Variable	Description
<code>wage</code>	Hourly wage (USD) — outcome
<code>experience</code>	Years of work experience
<code>education</code>	Years of education completed
<code>occupation</code>	Occupation type (worker / technical / services / office / sales / management)
<code>sector</code>	Industry sector (manufacturing / construction / other)

Before attempting the questions, run the setup chunk below. This loads the `AER` package, which contains the `CPS1985` dataset used in this lab.

```
#install.packages("tidyverse")
#install.packages("AER")
library(tidyverse)
library(AER)

data(CPS1985)
```

Questions

(i) Explore the data

Use `summary()` to inspect the dataset. Then create a boxplot of hourly wage by occupation. Which occupation has the highest median wage? Which has the lowest?

(ii) Simple linear regression

Fit a simple linear regression model with `wage` as the outcome and `experience` as the only predictor.

- Interpret the slope in context
- Use `confint()` to obtain a 95% confidence interval for the slope. Interpret it.
- What proportion of the variation in wages does experience alone explain?

(iii) Omitted variable bias

Add `education` to the model. How does the coefficient on `experience` change?

Explain this using the two omitted variable bias (OVB) conditions:

1. `education` is correlated with `wage` (use `cor()` to verify)
2. `education` is correlated with `experience` (use `cor()` to verify)

In which direction does omitting education bias the experience coefficient, and why?

(iv) Confidence intervals

Use `confint()` to obtain 95% confidence intervals for both coefficients in the model with `experience` and `education`.

- Interpret the CI for `experience` in context
- Interpret the CI for `education` in context
- What does it mean for a confidence interval to exclude zero?

(v) Confidence interval vs prediction interval

A new worker has 10 years of experience and 12 years of education. Use `predict()` with `interval = "confidence"` and `interval = "prediction"` to compute 95% intervals for their hourly wage.

- What is the point estimate?
- Compare the widths of the two intervals. Why is one wider than the other?

(vi) Interpolation vs extrapolation

Is 10 years of experience a safe basis for prediction? What about 60 years of experience? Check the observed range of `experience` in the dataset and discuss the risk of extrapolation.